

INTERNSHIP REPORT

PRESENTATION

Data Science Master

Virtual Internship

NITIN KUMAR

Roll No. 2410031504 | B.Tech CSE | Batch 2024–2028

DATA SCIENCE

PYTHON • PANDAS

NUMPY • EDA

VISUALIZATION

MACHINE LEARNING

ABOUT THE STUDENT

Candidate Profile

NAME

Nitin Kumar

ROLL NUMBER

2410031504

INSTITUTE

IILM University, Greater Noida, U.P.

PROGRAMME

B.Tech – Computer Science & Engineering

BATCH

2024–2028

INTERNSHIP DOMAIN

Siemens Data Science Master

DURATION

8 Weeks | April – June 2026

MODE

Virtual / Lab-based Learning

ABOUT THE PROGRAM

EduSkills, AICTE & Siemens Data Science

01

EduSkills Academy

Structured skill-development environment focused on practical learning and employability.

02

AICTE Patronage

Conducted under the AICTE–EduSkills Virtual Internship Program and National Internship Portal.

03

Learning Focus

Python, data handling, analysis, visualization, data quality, anomaly detection and ML concepts.

8 WEEKS • STRUCTURED LEARNING • HANDS-ON PRACTICE

PROBLEM STATEMENT

From Theory to Applied Data Science

Academic learning often explains Data Science, statistics and machine learning theoretically, while practical work with imperfect datasets requires a structured workflow.

01

DATA

Collect & load

02

CLEAN

Missing, duplicate &
invalid data

03

ANALYZE

EDA & statistics

04

DETECT

Anomalies & outliers

05

INSIGHT

Generate findings

PROJECT OBJECTIVES

What the Internship Set Out to Achieve

- Understand Data Science fundamentals and real-world applications.
- Develop practical Python skills for data analysis and manipulation.
- Use NumPy and Pandas for dataset handling and processing.
- Perform data cleaning and preprocessing.
- Conduct EDA to identify patterns, trends and relationships.
- Use visualization techniques to communicate findings.
- Apply statistics and identify outliers/anomalies.
- Understand the basic machine-learning workflow.

TECHNOLOGIES & TOOLS

The Toolkit Behind the Internship

01

Python

Core programming language

02

Pandas

Data manipulation & analysis

03

NumPy

Numerical computing

04

Matplotlib

Data visualization

05

Scikit-learn

Machine learning

06

Jupyter Notebook

Development environment

07

CSV / DataFrames

Dataset handling

08

Z-score

Anomaly analysis

SYSTEM ARCHITECTURE

From Raw Dataset to Meaningful Insights

01

DATA INPUT

CSV / structured datasets

02

DATA PROCESSING

Python + Pandas + NumPy

03

DATA QUALITY

Missing values • duplicates •
invalid values

04

ANALYSIS

EDA + statistical techniques

05

VISUALIZATION

Charts • patterns • trends

06

ANOMALY DETECTION

Outliers • Z-score

07

MACHINE LEARNING

Scikit-learn techniques

08

REPORTING

Findings & analytical results

METHODOLOGY

The 8-Week Learning Journey

1 Introduction to Data Science & Python

2 Data Loading & Dataset Profiling

3 Data Cleaning & Preprocessing

4 Generalized Data Quality Checks

5 Exploratory Data Analysis

6 Data Visualization & Statistical Analysis

7 Outlier & Anomaly Detection

8 Data Quality & Anomaly Detection Platform

FINAL: ASSESSMENT & PROJECT REVIEW

SCOPE & OUTCOMES

Skills and Capabilities Gained

SCOPE

- Complete Data Science workflow from fundamentals to analysis.
- Data collection, cleaning, preprocessing and profiling.
- EDA, visualization, statistical analysis and anomaly detection.
- Exposure to Python, NumPy, Pandas, Matplotlib and Scikit-learn.
- Virtual, simulation/lab-based learning environment.

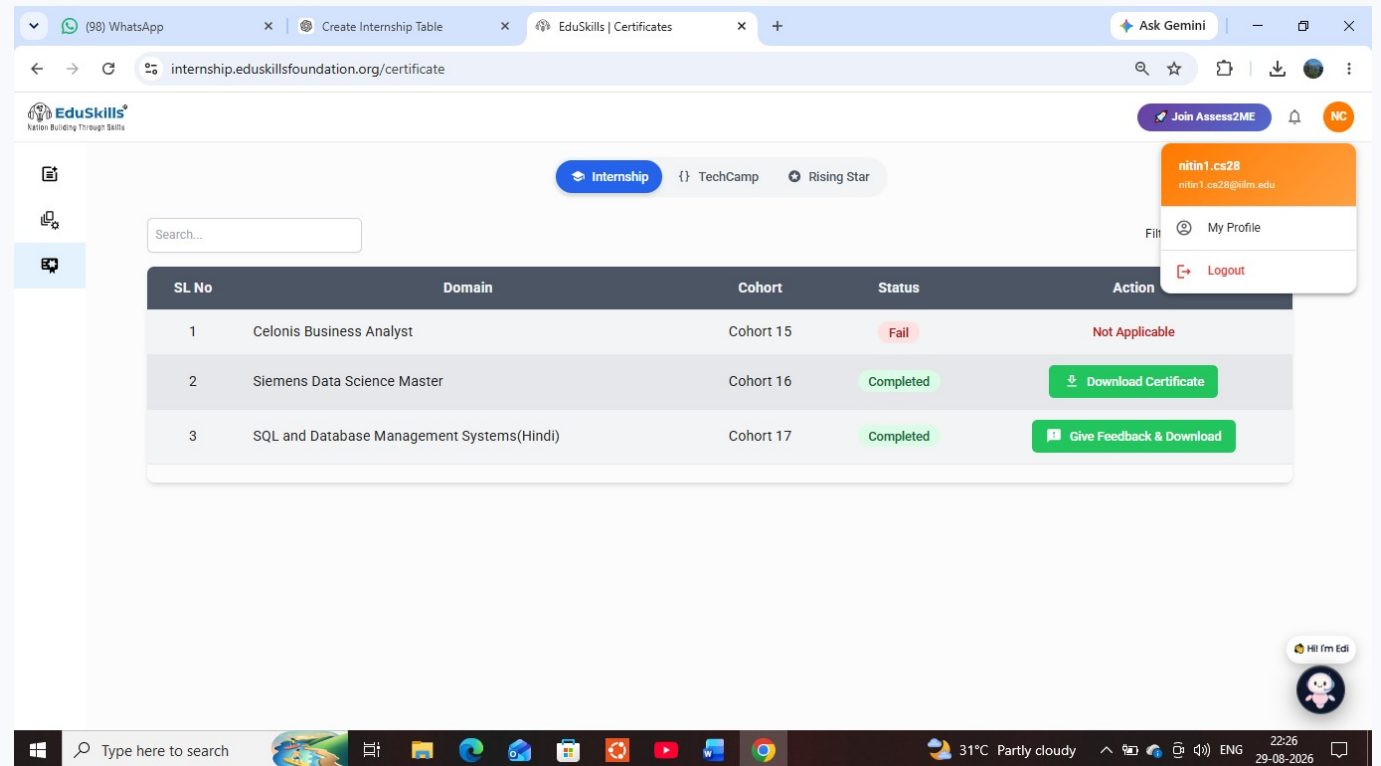
EXPECTED OUTCOMES

- Practical proficiency in Python, Pandas and NumPy.
- Ability to perform EDA and interpret patterns.
- Ability to visualize analytical findings.
- Identify outliers and anomalies using statistical techniques.
- Develop reusable data-quality checking functions.
- Build a structured Data Quality & Anomaly Detection workflow.

PRACTICAL EXPOSURE

Evidence of Internship Work

- Dataset handling
- Data-quality checks
- Analysis and visualization
- Anomaly detection
- Structured workflow



The screenshot displays the EduSkills Certificates portal in a web browser. The URL is internship.eduskillsfoundation.org/certificate. The page features a search bar and a table with the following data:

SL No	Domain	Cohort	Status	Action
1	Celonis Business Analyst	Cohort 15	Fail	Not Applicable
2	Siemens Data Science Master	Cohort 16	Completed	Download Certificate
3	SQL and Database Management Systems(Hindi)	Cohort 17	Completed	Give Feedback & Download

The interface includes a sidebar with navigation icons, a top navigation bar with 'Internship', 'TechCamp', and 'Rising Star' tabs, and a user profile dropdown for 'nitin1.cs28' with options for 'My Profile' and 'Logout'. The Windows taskbar at the bottom shows the date as 29-08-2026 and the time as 22:26.

Practical learning was completed in a virtual / simulation-based environment.

CERTIFICATION

Internship Completion



Ref No: C16Y2026M081400333

Internship Offer Letter

Student Name : Nitin Kumar
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : Siemens Data Science Master
Internship Duration : April 2026 - June 2026
AICTE Student ID : STU69a14c24d213e1772178468

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

Browser: Chrome | Downloaded on: 29-Aug-2026 03:17:38 PM IST

IP: 65.0.76.238 | Device: Desktop

Page 1



Certificate of Virtual Internship

This is to certify that

Nitin Kumar

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Science Master Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS

G.V. Pai

Gaurav Pai
Head - Strategy and Operations
Siemens Industry Software (India) Pvt. Ltd.

Dr. Buddha Chandrasekhar

Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev

Chief Executive Officer (CEO)
EduSkills



Certificate ID : 4bf4926f0d39a198fa0a

Student ID: STU69a14c24d213e1772178468



GRADE: O (Outstanding): 90-100 ; C (Excellent): 80-89 ; A (Very Good): 75-79 ; B (Good): 60-69 ; C (Fair): 50-59 ; D (Average): 40-49 ; P (Pass): 35-39 ; F (Fail): Below 35

THANK YOU

Data Science Master Virtual Internship

NITIN KUMAR | Roll No. 2410031504

IILM University, Greater Noida

AICTE – EduSkills Virtual Internship Program • Supported by Siemens